

CIS 665– Spring 2019

Final Exam – Part 2

You have two days to complete this exam. While taking the exam, you can refer to lecture materials provided for this class, HOEs that you have completed, notes you had personally taken during class sessions, and an ASP reference book or a website. Referring to or seeking help from any other source, human or otherwise, is explicitly prohibited. You should sequester or isolate yourself as best as you can when taking this exam.

Important note: If you violate the above rules or engage in any other form of academic misconduct, you will receive zero credit for this exam and one full letter grade deduction or an F grade for the course.

Use the three tables in your ExamDB (Final Exam – Part 1) to complete the exercises on this exam. The following tables are used as surrogates (you should have corresponding tables in your ExamDB), to provide examples for the exercises.

EMPLOYEE (**EmployeePK**, EmployeeName, EmployeeSalary, EmployeeHireDate, DepartmentFK)

This table has the foreign key and is referred to as **Table 1**

DEPARTMENT (**DepartmentPK**, DepartmentName)

This table is the source of the foreign key and is referred to as **Table 2**

USERINFO (**UserPK**, UserFirstName, UserLastName, UserLoginName, UserLoginPassword)

This table has user data and is referred to as **Table 3**

Create a new ASP.Net Core Web Application project within your ASP folder on your “W drive”. Call the project FinalExam, prefixed with the initials of your first and last names (e.g., Jane Doe would name her project, JDFinalExam).

Generate and write code to develop the titles, navigational structure, and functionalities illustrated in the sample exam project (<https://cisweb.biz.colostate.edu/CIS665/SP19Vijay.vijayasa/AspExamPublish>), and described below:

1) Search/filter records in Table 1 (15 points)

The search/filter criteria should include a) a wild card match on the values of a string column (e.g., EmployeeName), and b) a greater than or equal to match on the values of a numeric column (e.g., EmployeeSalary). Do not display the primary key (e.g., EmployeePK) or foreign key (e.g., DepartmentFK) in the results. Display the descriptive column (e.g., DepartmentName) associated with the foreign key (e.g., DepartmentFK) in the results. Include a link (i.e., Back to Full List”) to return to a display of all records from Table 1.

You can earn **10 bonus points** by adding the foreign key (e.g., DepartmentFK) as another search/filter criterion. This criterion should be implemented by giving users the option to select from a dynamically-generated drop-down list of items from Table 2. The list should display items from Table 2’s descriptive column (e.g., DepartmentName), and the values associated with the items should be Table 2’s primary key (e.g., DepartmentPK). In addition to the criteria on the string and numeric columns, the search/filter should satisfy this additional criterion: an equality match between the value of the foreign key column (e.g., DepartmentFK) and the primary key (e.g., DepartmentPK) of the item selected in the drop-down list.

2) Add/Insert records to Table 1 (10 points)

The drop-down list for the foreign key (e.g., DepartmentFK) should be dynamically generated. Include Required, MaxLength and RegularExpression validations and appropriate error messages for at least one string column (e.g., EmployeeName). Include Required and Range validations and appropriate error messages for a minimum of one numeric column (e.g., EmployeeSalary). After adding the new record, return the user to the Search/Filter view, and display a confirmation message.

3) Edit/Update the Profile of an authenticated user (20 points)

A user should be able to update his/her profile after logging in. Allow the user to modify only his/her first and last names (**Hint:** Make the username and password input controls “readonly”). After updating the user’s record, display a confirmation message. Include a Logout link button to log the user out, and return him/her to the Home view. **Note:** A user should not be able to view or modify any other user’s profile, besides their own.

You can test this functionality on the sample exam project by using these credentials – username: staples; password: nina.

Publish your project to AspExamPublish, and submit the URL.

Note: If any of your project files has a create/modify timestamp before or after the exam period, you will receive 0 credit for the exam.